main differences between a reversible and an irreversible reaction:

**Table 10.2** Differences between reversible and irreversible reactions

| REVERSIBLE REACTION | IRREVERSIBLE REACTION |
|---|---|
| It can be reversed under suitable conditions. | It cannot be reversed. |
| Both forward and backward reactions take place simultaneously. | It is unidirectional. It proceeds only in forward direction. |
| It attains equilibrium. | Equilibrium is not attained. |
| The reactants cannot be converted completely into products. | The reactants can be completely converted into products. |
| It is relatively slow. | It is fast. |

You will learn more about these reactions in your higher classes.

## 10.2 RATE OF A CHEMICAL REACTION

So far we discussed various types of chemical reactions and the nature of the reactants and products. Let us consider the following reactions:

- Rusting of iron
- Digestion of food
- Burning of petrol
- Weathering of rock

How fast is each reaction? Rank them from the slowest to fastest. How will you determine, which is the fastest and which is the slowest? One of the ways to find out how fast a reaction is as follows: Measure the amount of reactants or products before and after a specific period of time. For example, let us assume that 100 g of a substance 'A' undergoes a reaction and after an hour 50 g of 'A' is left.

$$A \rightarrow \text{Product}$$

In another instance, 100 g of substance 'C' undergoes a reaction and after an hour, 20 g of 'C' is left.

$$C \rightarrow \text{Product}$$

Can you say which is the faster reaction? In the first reaction, 50 g of the reactant (A) is converted into products whereas in the second reaction 80 g of the reactant is converted into products in one hour. So, the second reaction is faster. This measurement is called 'the reaction rate'.

"Rate of a reaction is the change in the amount or concentration of any one of the reactants or products per unit time".

Consider the following reaction

$$A \rightarrow B$$

The rate of this reaction is given by

$$\text{Rate} = -\frac{d[A]}{dt} = +\frac{d[B]}{dt}$$

Where,

[A] – Concentration of A

[B] – Concentration of B

The negative sign indicates the decrease in the concentration of A with time.

The postive sign indicates the increase in the concentration of B with time.

**Note:** '[ ]' represents the concentration, 'd' represents the infinitesimal change in the concentration.

### Why is reaction rate important?

Faster the reaction, more will be the amount of the product in a specified time. So, the rate of a reaction is important for a chemist for designing a process to get a good yield of a product. Rate of reaction is also important for a food processor who hopes to slow down the reactions that cause food to spoil.

### 10.2.1 Factors influencing the rate of a reaction

Can the rate of a reaction be changed? The rate of a reaction can be changed. For



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